Integration using Partial Fractions

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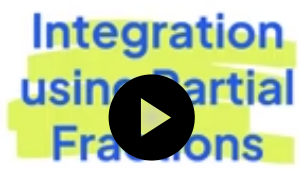

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Help me understand

Integration using partial fractions

What are partial fractions?

- This is the reverse process to adding (or subtracting) fractions
  - The common polynomial denominator is split into factors
- Make sure you are familiar with the notes on Partial Fractions

$$\frac{1}{x^2+3x+2} = \frac{1}{(x+2)(x+1)}$$

$$\frac{1}{(x+2)(x+1)} \equiv \frac{A}{(x+2)} + \frac{B}{(x+1)}$$

$$1 \equiv A(x+1) + B(x+2)$$

LET $x = -1$, $1 = B$, $B = 1$
LET $x = -2$, $1 = -A$, $A = -1$

$$\therefore \frac{1}{x^2+3x+2} = \frac{1}{x+1} - \frac{1}{x+2}$$

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Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Fractions with linear denominators can be integrated (See Integrating Other Functions)

$$\int \frac{1}{x+3}\,dx = \ln|x+3| + c$$

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- A fraction with a polynomial denominator (degree 2+) can be integrated by ...
  - ... splitting using partial fractions
  - ... integrating each partial fraction

e.g. $\displaystyle\int \frac{3}{x^2+5x+4}\,dx$

STEP 1: FACTORISE DENOMINATOR, IF NEEDED

$$\frac{3}{x^2+5x+4} = \frac{3}{(x+4)(x+1)}$$

STEP 2: EXPRESS AS PARTIAL FRACTIONS

$$\frac{3}{(x+4)(x+1)} \equiv \frac{A}{x+4} + \frac{B}{x+1}$$

$$3 \equiv A(x+1) + B(x+4)$$

LET $x = -1$, $3 = 3B$, $B = 1$

LET $x = -4$, $3 = -3A$, $A = -1$

$$\therefore \frac{3}{x^2+5x+4} = \frac{1}{x+1} - \frac{1}{x+4}$$

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STEP 3: INTEGRATE EACH TERM

$$\int \frac{3}{x^2+5x+4}\,dx = \int\left(\frac{1}{x+1} - \frac{1}{x+4}\right)dx$$

$$= \ln|x+1| - \ln|x+4| + c$$

REMEMBER MODULUS WHEN INTEGRATING TO ln

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- STEP 1: Factorise denominator, if needed
- STEP 2: Express as partial fractions
- STEP 3: 'Adjust' and 'compensate' each term so it can be integrated (See Reverse Chain Rule)
- STEP 4: Integrate each term (usually involves **ln**) and simplify

Examiner Tips and Tricks

- When there are several parts, read the whole question:
- An early part may be a "show that" involving partial fractions
- A later part may be to integrate the original fraction
- If you can't do, or get stuck on, the partial fractions bit of the question you can still use the "show that" result to

Worked Example

(a) Fully factorise $4x^3 - 4x^2 + x$

(b) Express $\dfrac{2}{4x^3 - 4x^2 + x}$ in partial fractions

(c) Find $\displaystyle\int \frac{2}{4x^3 - 4x^2 + x}\,dx$

a) $4x^3 - 4x^2 + x = x(4x^2 - 4x + 1)$

$= x(2x - 1)^2$

IF x IS NOT A FACTOR, FACTOR THEOREM AND / OR ALGEBRAIC DIVISION WOULD BE NEEDED TO FACTORISE A CUBIC

LOOKING AHEAD, THIS IS STEP 1 FOR PART c)

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